

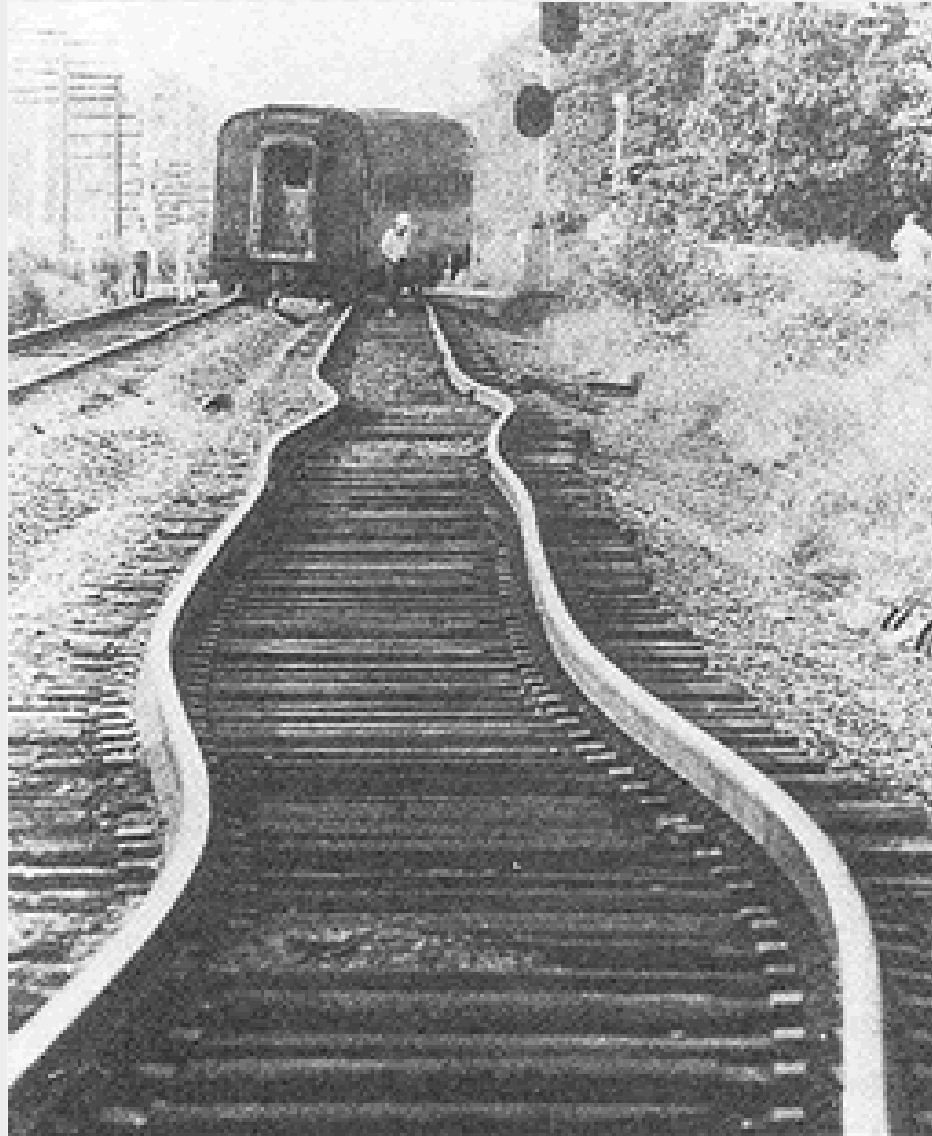
Physics: Section 12.5

Linear Expansion / Volumetric Expansion

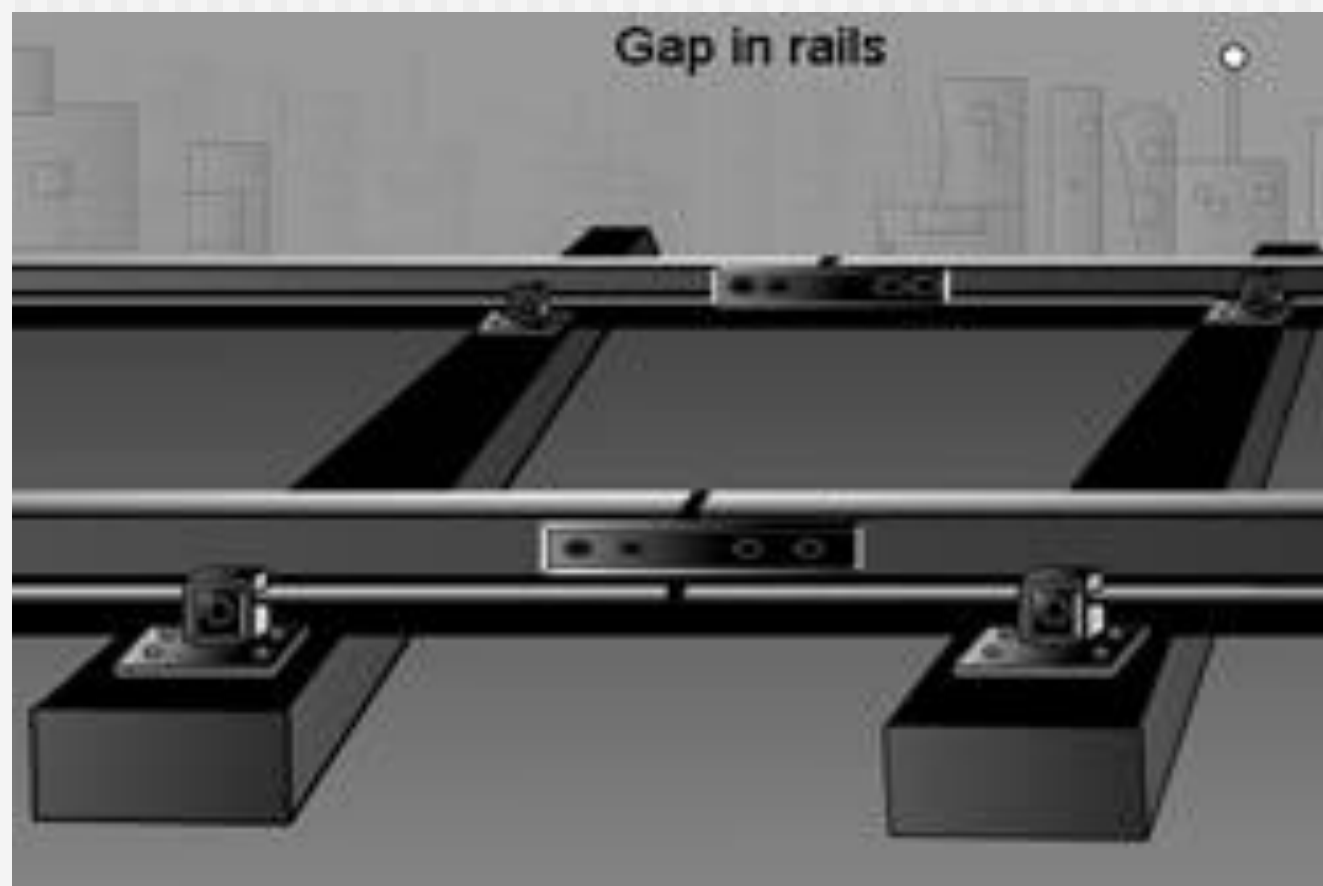


Thermal Expansion

- Most solids and liquids **expand when heated** and **contract when cooled**.
- In most solids and liquids, when temperature rises, their molecules vibrate more and have **more kinetic energy**.
- At these higher levels of vibration, the particles aren't bound to each other as tightly, **so the object expands**.







Thermal Expansion

- The amount an object expands can be calculated for both **one-dimensional (linear)** and **three-dimensional (volumetric)** expansion.
- The amount a material expands is characterized by the material's **coefficient of expansion**. Each substance has a **constant** for which describes its ability to expand.
- **Linear constants** are denoted by the Greek letter **alpha(α)**
- **Volumetric constants** are denoted by the Greek letter **beta (β)**

Thermal Expansion

Linear Expansion

Amount of linear expansion (ΔL) depends on the following:

1. $\Delta T (T_{\text{final}} - T_{\text{initial}})$: Change in temperature ($^{\circ}\text{C}$)
2. L_i : Original length
3. α (alpha) : Coefficient of thermal expansion

Units for Coefficient of Thermal Expansion

α units: $1/^{\circ}\text{C}$

Linear Expansion Formula

$$\Delta L = \alpha L_i \Delta T$$

Change
In
Length

Change
In
Temperature
(*Final – Initial*)

Original
Length

Coefficient
Of Thermal
Expansion

Linear Expansion Formula

$$\Delta L = \alpha L_i \Delta T$$



Linear Expansion - FYI

- **Alpha (α) is material dependent, i.e. it's different for steel, aluminum, concrete, etc.**
- **Since the coefficient of linear expansion is different for different materials, therefore each substance has its own rate of expansion.**
- **Reference Sheet**

Linear Expansion Example #1

A Brass strip is 3 cm long at 0°C. How long will it be at 100°C if the coefficient of linear expansion for Brass is $1.8 \times 10^{-5}/^{\circ}\text{C}$?

$$\Delta L = (\alpha)(L_i)(\Delta t)$$

$$\Delta L = (1.8 \times 10^{-5}/^{\circ}\text{C})(3 \text{ cm})(100^{\circ}\text{C} - 0^{\circ}\text{C})$$

$$\Delta L = 0.0054 \text{ cm}$$

$$L_f = \Delta L + L_i$$

$$L_f = 0.0054 \text{ cm} + 3 \text{ cm}$$

$$L_f = 3.0054 \text{ cm}$$

Linear Expansion Example #2

Metric rulers are calibrated at 20°C. What is the error in a measurement of 500 mm if made at 45°C. ($\alpha = 1.2 \times 10^{-5}/^{\circ}\text{C}$)

$$\Delta L = (\alpha)(L_i)(\Delta t)$$

$$\Delta L = (1.2 \times 10^{-5}/^{\circ}\text{C})(500 \text{ m})(45^{\circ}\text{C} - 20^{\circ}\text{C})$$

$$\Delta L = 0.15 \text{ mm}$$

Linear Expansion Example #3

A glass jar ($\alpha = 0.3 \times 10^{-5}/^{\circ}\text{C}$) has a metal lid ($\alpha = 1.6 \times 10^{-5}/^{\circ}\text{C}$) which is stuck. If you heat them by placing them in hot water, the lid will be:

- (A) Easier to open
- (B) Harder to open
- (C) Same

Linear Expansion Example #4

A brass rod is 0.70 m long at 40°C. Find the length of this rod at 50°C.

Linear Expansion Example #5

Compute the increase in length of 50 m of copper wire when its temperature changes from 12 to 32°C.

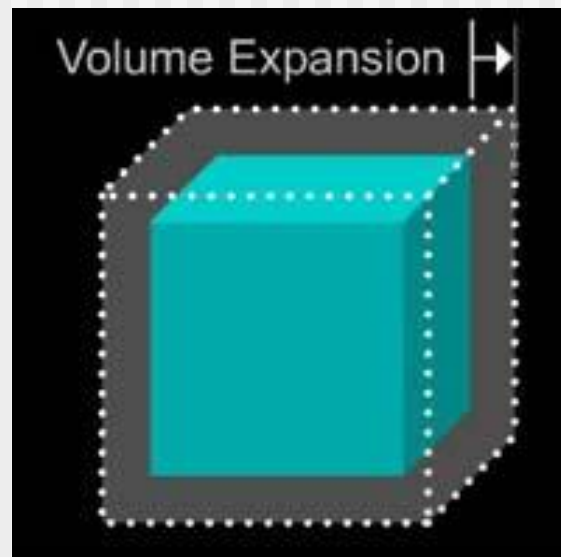
Linear Expansion Example #6

The diameter of a hole drilled through a piece of brass is 1.5 cm at 20.0°C. What is the diameter at 15.0°C?

J.S. Miller Video – Thermal Expansion

<http://www.youtube.com/watch?v=fZaAqRS6uOM>

Volumetric Expansion



Volumetric Expansion

Volumetric Expansion

Amount of volume expansion (ΔV) depends on the following:

1. $\Delta T (T_{\text{final}} - T_{\text{initial}})$: Change in temperature ($^{\circ}\text{C}$)
2. V_i : Original volume
3. Beta (β): Coefficient of volumetric expansion

Units for Coefficient of Volumetric Expansion:

β units: $1/^{\circ}\text{C}$

Volumetric Expansion Formula

$$\Delta V = \beta V_i \Delta T$$

Change
In
Volume

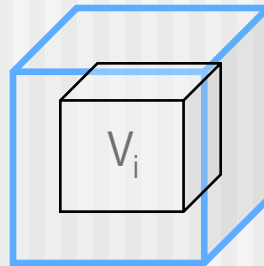
Change In
Temperature
Final - Initial

Original
Volume

Coefficient
Of Volumetric
Expansion

Volumetric Expansion Formula

$$\Delta V = \beta V_i \Delta T$$



$$V_i + \Delta V$$

Volumetric Thermal Expansion Example #1

The coefficient of volumetric expansion of mercury is $1.8 \times 10^{-4} / ^\circ\text{C}$. If the temperature of 10 L of mercury increases by $30 ^\circ\text{C}$, how much volume will the mercury increase by?

$$\Delta V = (\beta)(V_i)(\Delta t)$$

$$\Delta V = (1.8 \times 10^{-4})(10 \text{ L})(30 ^\circ \text{ C})$$

$$\Delta V = 0.054 \text{ L}$$

Volumetric Thermal Expansion Example #2

Suppose that the gas tank in your car is completely filled when the temperature is 17°C. How many gallons will spill out of the 20 gallon steel tank when the temperature rises to 35°C?

$$\beta_{gas} = 9.5 \times 10^{-4} / ^\circ C$$

$$\beta_{steel} = 0.36 \times 10^{-4} / ^\circ C$$

Volumetric Thermal Expansion Example #2

We need to find the difference from the expansion of the gas and the expansion of the steel:

$$\Delta V_{\text{gas}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{gas}} = (9.5 \times 10^{-4}/^{\circ}\text{C})(20 \text{ gal})(18^{\circ}\text{C})$$

$$\Delta V_{\text{gas}} = 0.342 \text{ gallons}$$

$$\Delta V_{\text{steel}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{steel}} = (0.36 \times 10^{-4}/^{\circ}\text{C})(20 \text{ gal})(18^{\circ}\text{C})$$

$$\Delta V_{\text{steel}} = 0.01296 \text{ gallons}$$

$$\Delta V_{\text{final}} = \Delta V_{\text{gas}} - \Delta V_{\text{steel}}$$

$$\Delta V_{\text{final}} = 0.342 \text{ gal} - 0.01296 \text{ gal}$$

$$\Delta V_{\text{final}} = 0.32904 \text{ gal}$$

Volumetric Thermal Expansion Example #3

The hot water heating system of a building contains 60.0 ft³ of water in steel pipes. How much water will overflow into the expansion tank if the system is filled at 40°C and then is heated to 80°C?

$$(\beta_{\text{steel}} = 0.36 \times 10^{-4} / ^\circ\text{C})$$

Volumetric Thermal Expansion Example #3:

We need to find the difference from the expansion of the water and the expansion of the steel:

$$\Delta V_{\text{water}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{water}} = (2.1 \times 10^{-4}/^{\circ}\text{C})(60 \text{ ft}^3)(40^{\circ} \text{ C})$$

$$\Delta V_{\text{water}} = 0.504 \text{ ft}^3$$

$$\Delta V_{\text{steel}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{steel}} = (0.36 \times 10^{-4}/^{\circ}\text{C})(60 \text{ ft}^3)(40^{\circ}\text{C})$$

$$\Delta V_{\text{steel}} = 0.0864 \text{ ft}^3$$

$$\Delta V_{\text{final}} = \Delta V_{\text{water}} - \Delta V_{\text{steel}}$$

$$\Delta V_{\text{final}} = 0.504 \text{ ft}^3 - 0.0864 \text{ ft}^3$$

$$\Delta V_{\text{final}} = 0.4176 \text{ ft}^3$$

Volumetric Thermal Expansion Example #4

A thermometer contains 0.50 cm³ of alcohol at room temperature (20°C) when Stanley takes it into the physics lab for an experiment. By how much does the volume of alcohol in the thermometer change after it sits in an 80°C Pyrex glass beaker?

$$(\beta_{\text{alcohol}} = 11.2 \times 10^{-4}/^{\circ}\text{C})$$

Volumetric Thermal Expansion Example #4:

We need to find the difference from the expansion of the alcohol and the expansion of the glass:

$$\Delta V_{\text{alcohol}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{alcohol}} = (11.2 \times 10^{-4}/^{\circ}\text{C})(.50 \text{ cm}^3)(60^{\circ}\text{C})$$

$$\Delta V_{\text{alcohol}} = 0.0336 \text{ cm}^3$$

$$\Delta V_{\text{glass}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{glass}} = (0.09 \times 10^{-4}/^{\circ}\text{C})(.50 \text{ cm}^3)(60^{\circ}\text{C})$$

$$\Delta V_{\text{glass}} = 0.00027 \text{ cm}^3$$

$$\Delta V_{\text{final}} = \Delta V_{\text{alcohol}} - \Delta V_{\text{glass}}$$

$$\Delta V_{\text{final}} = 0.0336 \text{ cm}^3 - 0.00027 \text{ cm}^3$$

$$\Delta V_{\text{final}} = 0.0333 \text{ cm}^3$$

Volumetric Thermal Expansion Example #5

A Pyrex glass of water with volume 1 liter is completely filled at 5°C. How much water will spill out of the Pyrex glass when the temperature is raised to 85°C?

Volumetric Thermal Expansion Example #5:

We need to find the difference from the expansion of the water and the expansion of the glass:

$$\Delta V_{\text{water}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{water}} = (2.1 \times 10^{-4}/^{\circ}\text{C})(1 \text{ L})(80^{\circ}\text{C})$$

$$\Delta V_{\text{water}} = 0.0168 \text{ L}$$

$$\Delta V_{\text{glass}} = (\beta)(V_i)(\Delta t)$$

$$\Delta V_{\text{glass}} = (0.09 \times 10^{-4}/^{\circ}\text{C})(1 \text{ L})(80^{\circ}\text{C})$$

$$\Delta V_{\text{glass}} = 0.00072 \text{ L}$$

$$\Delta V_{\text{final}} = \Delta V_{\text{water}} - \Delta V_{\text{glass}}$$

$$\Delta V_{\text{final}} = 0.0168 \text{ L} - 0.00072 \text{ L}$$

$$\Delta V_{\text{final}} = 0.01608 \text{ L}$$